Maciej Tkacz

Junior Robotics Engineer - Prototyping & Simulation

Email: maciek01110@gmail.com | Phone: +48 881 912 125 | Location: Krakow, Poland | Polish: native | English: C1 | Driving licence: category B

PROFESSIONAL PROFILE

Hands-on robotics builder: Mechanical Engineering student at Cracow University of Technology and certified Software Technician who takes electromechanical systems from concept to working prototype. Commercial experience creating robotics simulations and digital twins in Visual Components for automotive production. Independently designed and built a SCARA robotic arm - CAD, 3D-printed structure, NEMA stepper actuators with TMC/DRV drivers, microcontroller electronics and Python/C++ control software. Pragmatic, fast-iteration prototyping mindset and strong interest in physical AI and robotics automation. Fluent Polish, English C1; available full-time thanks to a flexible study schedule.

KEY SKILLS

- Hands-on prototyping & hardware: end-to-end electromechanical builds - CAD design (SolidWorks, Inventor, Fusion 360), 3D printing (FDM/SLA/MJF), component sourcing and selection (NEMA steppers, TMC/DRV drivers, microcontrollers), assembly, wiring and bring-up testing.
- Mechatronics & control: kinematics, control systems, dynamic systems modelling, automation and mechatronics coursework; industrial robot programming (Kawasaki, teach pendant).
- Programming: Python, C++, Arduino/C; hardware-control code, automation and calculation scripts, iterative debugging of electromechanical prototypes.
- Robotics software & simulation: Visual Components digital twins (robot paths, collision checks, cycle flows, layout optimization); ROS self-study; MATLAB at university; simulation-to-real validation mindset.
- Rapid iteration & ownership: personal R&D projects taken from concept to working prototype; reverse engineering; technical documentation; pragmatic build-test-improve approach.
- Collaboration: fluent Polish (native) and English (C1); teamwork in interdisciplinary engineering environments.

SELECTED ROBOTICS PROJECT

SCARA Robotic Arm Prototype | Personal R&D Project

- Designed and built an end-to-end SCARA robotic arm prototype, covering mechanical concept, CAD modelling, kinematic layout, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Optimized components for 3D printing, including PET-G and carbon-fibre reinforced materials, with focus on stiffness, assembly constraints and rapid iteration.
- Developed Python/C++ control software for motion and hardware-interaction experiments, applying forward/inverse kinematics and control-system fundamentals.
- Integrated mechanical, electronic and software subsystems through iterative assembly, testing and debugging, gaining practical hardware-software integration experience.

PROFESSIONAL EXPERIENCE

Application Engineer | AIAutomation | Jan 2025 - May 2026

- Created robotics simulations and digital twins of production workcells in Visual Components for automotive clients.
- Developed robot logic, motion sequences, robot paths, collision checks, cycle flows and virtual process validation.
- Optimized workcell layouts and 3D geometry for simulation needs, using SolidWorks and Autodesk Inventor.
- Analyzed customer documentation, engineering standards and technical specifications to support solution concepts aligned with production requirements.
- Collaborated with engineers and stakeholders, communicating simulation assumptions, constraints and improvement proposals.

3D Printing and CAD Design Specialist | Cubic Inch Additive Manufacturing, Piaseczno | Jun 2023 - Aug 2023

- Operated and serviced FDM, MJF and SLA 3D printers, supervising process parameters, post-processing and quality control.
- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing and rapid prototyping.
- Supported implementation testing for a new SLA technology, documenting progress and technical observations.
- Coordinated production tasks in a 10-person project team, balancing quality, manufacturability and delivery constraints.

Robotics and 3D Printing Intern | ASTOR Robotics Center, Krakow | May 2022

- Assembled mechanical equipment and 3D-printed components for Kawasaki robots and Astorino educational robot platforms.
- Programmed Kawasaki industrial robots using teach pendant workflows and created basic robot motion programs.
- Tested robot and workstation operation, gaining practical exposure to robot setup, safety and hardware integration.

Web Application Development Intern | Souczek Design Studio Reklamy i Druku, Kielce | Jul 2021

- Solved technical problems in JavaScript, including implementation of an interactive order form.
- Tested and deployed web functionality in a team environment according to customer guidelines.

EDUCATION

- Cracow University of Technology - Mechanical Engineering, Engineer's degree in progress (Oct 2024 - present). Relevant coursework: dynamic systems modelling, automation/control, analytical mechanics and mechatronics.
- PKMechPower Student Research Group, Mechanical Section - CAD and 3D printing tasks, including driver seat and brake-system components; mass and strength-oriented design optimization.
- Zespol Szkol Informatycznych im. gen. Jozefa Hauke-Bosaka, Kielce - Software Technician (Sep 2019 - Apr 2024).

CERTIFICATES

- Kawasaki robot operation and programming - integrator course, ASTOR Robotics Center (May 2022).
- Python programming courses and self-directed Python/C++ development in robotics and automation projects.